

PROJECT PROPOSAL: PROJECT ŚIPRA

VALUE-ENGINEERED AUTONOMOUS ACTIVE-AIRBAG COMMUTER HELMET

Document Type:	Technical & Economic Feasibility Concept Note
Target Market:	High-Volume Two-Wheeler Commuter Safety (Mass Market)
System Architecture:	Multi-Modal AI Sensor Fusion Predictive Ecosystem (App-Free)

1. Executive Summary

Project Śipra addresses a primary market gap in global two-wheeler safety: providing proactive cervical spine protection to the everyday commuter at an accessible price point. Historically, active airbag systems have been confined to premium racing vests or high-end apparel costing hundreds of dollars. Project Śipra breaks this cost barrier by optimizing the delicate balance between advanced machine-vision intelligence and highly abundant, mass-produced consumer pneumatic components.

To eliminate the industry's two most dangerous technical risks—false positives (accidental deployment during aggressive riding) and false negatives (delayed triggering during a crash)—this updated architecture introduces a ****Multi-Modal AI Sensor Fusion Pipeline****. By running a localized fusion model directly on affordable edge hardware, the system reconciles visual optical stream matrices with kinematic telemetry. This self-contained, app-free device ensures absolute activation precision within milliseconds, priced at just a small fraction higher than a standard commuter helmet.

2. System Architecture & Multi-Modal Sensor Fusion

The core breakthrough of Project Śipra v3 lies in its validation layer. Rather than relying solely on one data medium, the onboard edge processor runs a real-time **Kinematic-Visual Fusion Model**. By cross-verifying what the helmet "sees" with what the helmet "feels," false positives caused by rapid intentional head-checks, extreme potholes, or looking down are entirely filtered out.

2.1. The Sensor Stream Ingestion Matrix

- **Optical Module (Camera + Optical Flow):** A low-cost, fixed-focus micro-camera tracks continuous horizon lines and pixel structural arrays. It runs an optimized *Optical Flow* algorithm calculating *Time-to-Collision (TTC)* vectors. If the bike slides out, the camera immediately flags an abnormal environment angular tilt.
- **Inertial Module (IMU + Gyroscope):** A standard, highly responsive 6-axis Inertial Measurement Unit (incorporating a 3-axis accelerometer and a 3-axis gyroscope) continuously maps linear acceleration, G-forces, and angular velocity rates across the X, Y, and Z axes.

2.2. The AI Fusion and Trigger Logic

The telemetry streams merge locally at the micro-board level via a lightweight **Complementary Kalman Filter / Thresholding Machine Learning Network**. Deployment commands are strictly gated by two-tier synchronization:

- 1. Scenario A (The Pothole / Aggressive Head Turn):** The IMU registers a high G-force shock spike or the Gyroscope notes a sudden rotation. However, the camera's Optical Flow shows a stable, predictable horizon path ahead. The AI Fusion Model registers a *Mismatched Matrix* and ****suppresses deployment****, preventing a costly false positive.
- 2. Scenario B (True Accident State / High-Side or Low-Side Fall):** The IMU detects free-fall weightlessness or violent vector deceleration, the Gyroscope notes uncontrolled rolling velocity, and the Optical Flow simultaneously confirms a chaotic environmental inversion matrix. Both streams converge with high confidence. The AI Fusion Model registers an **Active Trigger Match** and fires the solenoids immediately within $\leq 2ms$ of calculation time.

3. Component Breakdown & Bill of Materials (BOM)

Integrating an IMU sensor along with the camera module adds negligible overhead, as 6-axis motion chips are mass-manufactured at massive consumer scales for smartphones and IoT gadgets:

COMPONENT MODULE	TECHNICAL SPECIFICATIONS	WHOLESALE COST (USD)	WHOLESALE COST (INR)
Edge-AI Fusion Board	ESP32-S3 AI processor + 2MP CMOS lens + Integrated 6-Axis IMU (Acc/Gyro)	\$9.50 – \$13.50	₹780 – ₹1,100
Pneumatic Cartridges	2× Threaded consumer \$CO_2\$ or \$N_2\$ canisters (Bicycle standard)	\$3.00 – \$5.00	₹250 – ₹400
Solenoid Puncture Valves	Low-resistance, high-speed electromagnetic pins	\$6.00 – \$8.00	₹500 – ₹650
Deployment Airbag	Automated die-cut, heat-sealed TPU nylon bladder	\$4.00 – \$6.00	₹330 – ₹500
Power Supply System	1,000 mAh Li-Po pouch cell + Type-C charging port + Wake Sensor	\$3.50 – \$4.50	₹290 – ₹370
TOTAL MODULE COST	Raw manufacturing cost of multi-modal safety core	\$26.00 – \$37.00	₹2,150 – ₹3,020

Projected Mass Market Retail Pricing

Even with the multi-sensor upgrade, the consumer pricing brackets remain firmly inside the targeted value-engineered threshold:

- Standard Safe Commuter Helmet Baseline Retail: ₹1,500 – ₹2,000 INR
- Project Śipra Integrated AI Sensor Fusion Upgrade: + ₹3,200 – ₹3,700 INR
- Target Retail Consumer Price Point: ₹4,700 – ₹5,700 INR (\$57 – \$68 USD)

4. Power Optimization Profile

- **Deep Sleep State:** When the helmet is stationary, the entire core (Processor, Camera, IMU) is fully powered down. The micro-vibration sensor sips negligible micro-amps. The helmet can sit dormant for **6 to 12 months** without exhausting its battery.
- **Active Commuting State:** Physical handling instantly boots the fusion pipeline. Running the sensor fusion model and frame processing simultaneously draws roughly 120–160 mA. The 1,000 mAh battery yields ****6 to 8 hours** of continuous, active multi-modal road tracking**. For an average commuter riding 1 hour daily, the device requires charging only ****once a week**** via a standard Type-C mobile cable.

5. User Experience & Maintenance Paradigm

Project Śipra eliminates smartphone applications, pairing routines, and wireless handshakes to achieve 100% fail-safe standalone autonomy.

- **Frictionless Operation:** The rider treats it like a regular helmet. Putting it on automatically arms the sensor fusion ecosystem. A low-profile indicator LED placed near the bottom rim glows a solid **Green** in the user's peripheral vision, confirming multi-sensor calibration is live and ready.
- **Intuitive Onboard Diagnostics:** Status mapping is handled locally without complex user screens:
 - Green:* All sensors calibrated, armed, and actively tracking.
 - Flashing Yellow + Audio Beep:* Low battery alert (indicating a weekend charge is required).
 - Red + Continuous Tone:* Sensor fault or spent pneumatic cartridge.
- **Post-Deployment Maintenance:** If an accident occurs, the rigid air pillow inflates downward and inward within 25–35ms, cushioning the neck and transferring the kinetic load safely to the shoulders. The primary electronics and sensors remain structurally unharmed. The user simply pulls back the cheek padding, unscrews the spent cylinders, and screws in two new standard bicycle cartridges at a negligible cost.